

Symbol	Code Name	Parameter Definition	Implemented In	Declared In	Department
INPUTS					
<i>n_{w,tot}</i>	MB.m_tot	Wing structural mass (kg)	StructuresAll.m	MassBuildup.m	Auto-computed
<i>M_{TOM}</i> (prev)	adp.MTOM (current)	MTOM from previous iteration (kg)	StructuresAll.m	ADP.m	Mass & Stability
<i>b</i> (prev)	adp.Span (current)	Wingspan from previous iteration (m)	StructuresAll.m	ADP.m	Aerodynamics

Symbol	Code	Parameter Definition	Implemented In	Declared In	Department
INPUTS					
<i>N</i>	G.N	Number of spanwise	StiffnessDistribution.m	WingGeometry.m	Auto-computed
<i>L_h</i>	G.L.hinge	Hinge station index	StiffnessDistribution.m	WingGeometry.m	Auto-computed
<i>h_{yl}</i>	G.h_wb(l)	Wingbox height at station l	StiffnessDistribution.m	WingGeometry.m	Auto-computed
<i>w_{yl}</i>	G.w_wb(l)	Wingbox width at station l	StiffnessDistribution.m	WingGeometry.m	Auto-computed
<i>A_{encly}</i>	G.A _{enc} (l)	Enclosed box area at station l	StiffnessDistribution.m	WingGeometry.m	Auto-computed
<i>t_{sk}</i>	W.t _{sk} (l)	Skin thickness at station l	StiffnessDistribution.m	WingboxSizing.m	Auto-computed
<i>A_{caply}</i>	W.A _{cap} (l)	Spar cap area at station l	StiffnessDistribution.m	WingboxSizing.m	Auto-computed
<i>E</i>	W.E _{mat}	Young's modulus (Pa)	StiffnessDistribution.m	WingboxSizing.m	Auto-computed
<i>G</i>	W.G _{mat}	Shear modulus (Pa)	StiffnessDistribution.m	WingboxSizing.m	Auto-computed

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<i>N</i>	G.N	Number of spanwise	StiffnessDistribution.m	WingGeometry.m	Auto-computed
<i>L_h</i>	G.L.hinge	Hinge station index	StiffnessDistribution.m	WingGeometry.m	Auto-computed
<i>h_{yl}</i>	G.h_wb(l)	Wingbox height at station l	StiffnessDistribution.m	WingGeometry.m	Auto-computed
<i>w_{yl}</i>	G.w_wb(l)	Wingbox width at station l	StiffnessDistribution.m	WingGeometry.m	Auto-computed
<i>A_{encly}</i>	G.A _{enc} (l)	Enclosed box area at station l	StiffnessDistribution.m	WingGeometry.m	Auto-computed
<i>t_{sk}</i>	W.t _{sk} (l)	Skin thickness at station l	StiffnessDistribution.m	WingboxSizing.m	Auto-computed
<i>A_{caply}</i>	W.A _{cap} (l)	Spar cap area at station l	StiffnessDistribution.m	WingboxSizing.m	Auto-computed
<i>E</i>	W.E _{mat}	Young's modulus (Pa)	StiffnessDistribution.m	WingboxSizing.m	Auto-computed
<i>G</i>	W.G _{mat}	Shear modulus (Pa)	StiffnessDistribution.m	WingboxSizing.m	Auto-computed

Symbol	Code Name	Parameter Definition	Implemented In	Declared In	Department
INPUTS					
<i>N</i>	G.N	Number of stations	WingboxSizing.m	WingGeometry.m	Auto-computed
<i>L_h</i>	G.L.hinge	Hinge station index	WingboxSizing.m	WingGeometry.m	Auto-computed
<i>h_{yl}</i>	G.h_wb(l)	Wingbox height at station l (m)	WingboxSizing.m	WingGeometry.m	Auto-computed
<i>w_{yl}</i>	G.w_wb(l)	Wingbox width at station l (m)	WingboxSizing.m	WingGeometry.m	Auto-computed
<i>A_{encly}</i>	G.A _{enc} (l)	Enclosed box area at station l (m ²)	WingboxSizing.m	WingGeometry.m	Auto-computed
<i>M_{yl}</i>	S.M(l)	Bending moment at station l (Nm)	WingboxSizing.m	SMT.m	Auto-computed
<i>Q_{yl}</i>	S.Q(l)	Shear force at station l (N)	WingboxSizing.m	SMT.m	Auto-computed
<i>T_{yl}</i>	S.T(l)	Torque at station l (Nm)	WingboxSizing.m	SMT.m	Auto-computed
<i>E_{AI}</i>	loc.AI.E = 71 GPa	AI 7075-T6 Young's modulus (Pa)	WingboxSizing.m	AircraftParams.m	Wing Structures
<i>G_{AI}</i>	loc.AI.G = 26.9 GPa	AI 7075-T6 shear modulus (Pa)	WingboxSizing.m	AircraftParams.m	Wing Structures
<i>nu_{AI}</i>	loc.AI.nu = 0.33	AI Poisson's ratio	WingboxSizing.m	AircraftParams.m	Wing Structures
<i>rho_{AI}</i>	loc.AI.rho = 2780	AI density (kg/m ³)	WingboxSizing.m	AircraftParams.m	Wing Structures
<i>sig_{AI}</i>	loc.AI.sig _{all} = 275 MPa	AI allowable stress (Pa)	WingboxSizing.m	AircraftParams.m	Wing Structures
<i>tau_{AI}</i>	loc.AI.tau _{all} = 155 MPa	AI allowable shear stress (Pa)	WingboxSizing.m	AircraftParams.m	Wing Structures
<i>L_{min,AI}</i>	loc.AI.L _{min} = 2 mm	AI minimum skin gauge (m)	WingboxSizing.m	AircraftParams.m	Wing Structures
<i>tw_{min,AI}</i>	loc.AI.L _{web} _{min} = 3	AI minimum web gauge (m)	WingboxSizing.m	AircraftParams.m	Wing Structures
<i>Ac_{min,AI}</i>	loc.AI.A _{cap} _{min} = 1e-4	AI minimum spar cap area (m ²)	WingboxSizing.m	AircraftParams.m	Wing Structures
<i>G_{CF}</i>	loc.CF.E = 59.8 GPa	CFRP Young's modulus (E8 rule) (Pa)	WingboxSizing.m	AircraftParams.m	Wing Structures
<i>G_{CF}</i>	loc.CF.G = 18.8 GPa	CFRP shear modulus (Pa)	WingboxSizing.m	AircraftParams.m	Wing Structures
<i>nu_{CF}</i>	loc.CF.nu = 0.33	CFRP Poisson's ratio	WingboxSizing.m	AircraftParams.m	Wing Structures
<i>rho_{CF}</i>	loc.CF.rho = 1550	CFRP density (kg/m ³)	WingboxSizing.m	AircraftParams.m	Wing Structures
<i>sig_{CF}</i>	loc.CF.sig _{all} = 220 MPa	CFRP allowable (post-BVID) (Pa)	WingboxSizing.m	AircraftParams.m	Wing Structures
<i>tau_{CF}</i>	loc.CF.tau _{all} = 110 MPa	CFRP allowable shear (Pa)	WingboxSizing.m	AircraftParams.m	Wing Structures
<i>b_{rib}</i>	loc.rib.spacing = 0.60	Rib pitch / skin panel width (m)	WingboxSizing.m	AircraftParams.m	Wing Structures
<i>k</i>	loc.k.buckle = 4.0	Simply-supported plate buckling coeff.	WingboxSizing.m	AircraftParams.m	Wing Structures

Symbol	Code Name	Parameter Definition	Implemented In	Declared In	Department	I/O
INPUTS						
<i>b</i>	adp.Span	Flight wingspan (m)	WingGeometry.m	ADP.m	Aerodynamics	INPUT
<i>S</i>	adp.WingArea	Wing reference area (m ²)	WingGeometry.m	ADP.m	Aerodynamics	INPUT
<i>N</i>	loc.N.stations =	Number of spanwise	WingGeometry.m	AircraftParams.m	Wing Structures	INPUT
<i>lambdab</i>	loc.lambdab = 0.30	Taper ratio b_1/b_2	WingGeometry.m	AircraftParams.m	Aerodynamics	INPUT
<i>L_{em,LE}</i>	loc.sweep.LE.deg	Leading edge sweep angle	WingGeometry.m	AircraftParams.m	Aerodynamics	INPUT
<i>L_{em,CE}</i>	loc.sweep.CE.deg	Half-chord sweep angle	WingGeometry.m	AircraftParams.m	Aerodynamics	INPUT
<i>(t/c)_r</i>	loc.t/c.root = 0.150	Thickness/chord at root	WingGeometry.m	AircraftParams.m	Aerodynamics	INPUT
<i>(t/c)_l</i>	loc.t/c.tip = 0.10	Thickness/chord at tip	WingGeometry.m	AircraftParams.m	Aerodynamics	INPUT
<i>f_{s, fwd}</i>	loc.f _s .fwd = 0.15	Front spar position / chord	WingGeometry.m	AircraftParams.m	Wing Structures	INPUT
<i>f_{s, aft}</i>	loc.f _s .aft = 0.60	Rear spar position / chord	WingGeometry.m	AircraftParams.m	Wing Structures	INPUT
<i>k_h</i>	loc.k _h .hinge = 32.5	Fold hinge spanwise	WingGeometry.m	AircraftParams.m	Wing Structures	INPUT
<i>N_{en}</i>	C.N _{en} = 2	Number of engines	WingGeometry.m	geom.constants	Wing Structures	INPUT

Symbol	Code Name	Parameter Definition	Implemented In	Declared In	Department
INPUTS					
<i>M_{TOM}</i>	adp.MTOM	Max take-off mass	EmpiricalMass.m	ADP.m	Mass & Stability
<i>b</i>	adp.Span	Wingspan (m)	EmpiricalMass.m	ADP.m	Aerodynamics
<i>S</i>	adp.WingArea	Wing reference area	EmpiricalMass.m	ADP.m	Aerodynamics
<i>M_F</i>	adp.MF _{Fuel}	Fuel mass fraction	EmpiricalMass.m	ADP.m	Perf / Propulsion
<i>M_{F,Res}</i>	adp.MF _{Res}	Reserve fuel fraction	EmpiricalMass.m	ADP.m	Perf / Propulsion
<i>g</i>	loc.g	Gravitational	EmpiricalMass.m	cast/atmos.m	Perf / Propulsion
<i>G₀</i>	loc.atmoslitur.Alt.cruise	Speed of sound at	EmpiricalMass.m	cast/atmos.m	Perf / Propulsion
<i>G₀</i>	loc.S ₀ = 2.0346	kg to lb unit	EmpiricalMass.m	SI.m	Shared / MDO
<i>G₀</i>	loc.S ₀ = 10.764	m ² to ft ² unit	EmpiricalMass.m	SI.m	Shared / MDO
<i>sf</i>	loc.sf = 1.5	Positive limit load	EmpiricalMass.m	AircraftParams.m	Regulatory
<i>sf</i>	loc.sf = 1.5	Safety factor limit to	EmpiricalMass.m	AircraftParams.m	Regulatory
<i>lambdab</i>	loc.lambdab	Taper ratio	EmpiricalMass.m	AircraftParams.m	Aerodynamics
<i>(t/c)_r</i>	loc.t/c.root	Root t/c ratio	EmpiricalMass.m	AircraftParams.m	Aerodynamics
<i>L_{em,CE}</i>	loc.sweep.CE.deg	Quarter-chord sweep	EmpiricalMass.m	AircraftParams.m	Aerodynamics
<i>L_{em,CE}</i>	loc.sweep.CE.deg	Half-chord sweep	EmpiricalMass.m	AircraftParams.m	Aerodynamics
<i>sig_{AI}</i>	loc.AI.sig _{all} = 275 MPa	AI 7075-T6 allowable	EmpiricalMass.m	AircraftParams.m	Wing Structures
<i>rho_{AI}</i>	loc.AI.rho = 2780	AI 7075-T6 density	EmpiricalMass.m	AircraftParams.m	Wing Structures
<i>sig_{CF}</i>	loc.CF.sig _{all} = 220 MPa	CFRP allowable	EmpiricalMass.m	AircraftParams.m	Wing Structures
<i>rho_{CF}</i>	loc.CF.rho = 1550	CFRP density (kg/m ³)	EmpiricalMass.m	AircraftParams.m	Wing Structures
<i>k_h</i>	loc.k _h .hinge	Fold hinge position	EmpiricalMass.m	AircraftParams.m	Wing Structures
<i>Z_h</i>	loc.z _h .hinge_mech = 0.10	Hinge mechanism	EmpiricalMass.m	AircraftParams.m	Wing Structures
<i>m_{h,min}</i>	loc.m _h .min = 1500	Minimum hinge mass	EmpiricalMass.m	AircraftParams.m	Wing Structures
<i>MR</i>	G.MR	Aspect ratio	EmpiricalMass.m	WingGeometry.m	Auto-computed
<i>b_{c,r}</i>	G.b _{c,r}	Root chord (m)	EmpiricalMass.m	WingGeometry.m	Auto-computed

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INPUTS					
<i>M_{TOM}</i>	adp.MTOM	Max take-off mass (kg)	LoadDistribution.m	ADP.m	Mass & Stability
<i>b</i>	adp.Span	Wingspan (m)	LoadDistribution.m	ADP.m	Aerodynamics
<i>M_F</i>	adp.MF _{Fuel}	Fuel mass fraction	LoadDistribution.m	ADP.m	Perf / Propulsion
<i>M_{F,Res}</i>	adp.MF _{Res}	Reserve fuel fraction	LoadDistribution.m	ADP.m	Perf / Propulsion
<i>g</i>	loc.g	Gravitational	LoadDistribution.m	SI.m	Shared / MDO
<i>G₀</i>	loc.S ₀ = 2.0346	kg to lb unit	LoadDistribution.m	SI.m	Shared / MDO
<i>G₀</i>	loc.S ₀ = 10.764	m ² to ft ² unit	LoadDistribution.m	SI.m	Shared / MDO
<i>sf</i>	loc.sf = 1.5	Positive limit load	LoadDistribution.m	AircraftParams.m	Regulatory
<i>sf</i>	loc.sf = 1.5	Safety factor CS-25.303	LoadDistribution.m	AircraftParams.m	Regulatory
<i>lambdab</i>	loc.lambdab	Taper ratio	LoadDistribution.m	AircraftParams.m	Wing Structures
<i>(t/c)_r</i>	loc.t/c.root	Root t/c ratio	LoadDistribution.m	AircraftParams.m	Wing Structures
<i>L_{em,CE}</i>	loc.sweep.CE.deg	Quarter-chord sweep	LoadDistribution.m	AircraftParams.m	Wing Structures
<i>L_{em,CE}</i>	loc.sweep.CE.deg	Half-chord sweep	LoadDistribution.m	AircraftParams.m	Wing Structures
<i>sig_{AI}</i>	loc.AI.sig _{all} = 275 MPa	AI 7075-T6 allowable	LoadDistribution.m	AircraftParams.m	Wing Structures
<i>rho_{AI}</i>	loc.AI.rho = 2780	AI 7075-T6 density	LoadDistribution.m	AircraftParams.m	Wing Structures
<i>sig_{CF}</i>	loc.CF.sig _{all} = 220 MPa	CFRP allowable	LoadDistribution.m	AircraftParams.m	Wing Structures
<i>rho_{CF}</i>	loc.CF.rho = 1550	CFRP density (kg/m ³)	LoadDistribution.m	AircraftParams.m	Wing Structures
<i>k_h</i>	loc.k _h .hinge	Fold hinge position	LoadDistribution.m	AircraftParams.m	Wing Structures
<i>Z_h</i>	loc.z _h .hinge_mech = 0.10	Hinge mechanism	LoadDistribution.m	AircraftParams.m	Wing Structures
<i>m_{h,min}</i>	loc.m _h .min = 1500	Minimum hinge mass	LoadDistribution.m	AircraftParams.m	Wing Structures
<i>MR</i>	G.MR	Aspect ratio	LoadDistribution.m	WingGeometry.m	Auto-computed
<i>b_{c,r}</i>	G.b _{c,r}	Root chord (m)	LoadDistribution.m	WingGeometry.m	Auto-computed

Symbol	Code Name	Parameter Definition	Implemented In	Declared In	Department
INPUTS					
<i>N</i>	G.N	Number of spanwise stations	SMT.m	WingGeometry.m	Auto-computed
<i>gy</i>	G.dy	Station spacing (m)	SMT.m	WingGeometry.m	Auto-computed
<i>L_h</i>	G.L.hinge	Hinge station index	SMT.m	WingGeometry.m	Auto-computed
<i>h_{yl}</i>	G.h_wb(l)	Wingbox height at station l (m)	SMT.m	WingGeometry.m	Auto-computed
<i>w_{yl}</i>	G.w_wb(l)	Wingbox width at station l (m)	SMT.m	WingGeometry.m	Auto-computed
<i>A_{encly}</i>	G.A _{enc} (l)	Enclosed box area at station l (m ²)	SMT.m	WingGeometry.m	Auto-computed
<i>t_{sk}</i>	W.t _{sk} (l)	Skin thickness at station l	SMT.m	WingboxSizing.m	Auto-computed
<i>A_{caply}</i>	W.A _{cap} (l)	Spar cap area at station l	SMT.m	WingboxSizing.m	Auto-computed
<i>E</i>	W.E _{mat}	Young's modulus (Pa)	SMT.m	WingboxSizing.m	Auto-computed
<i>G</i>	W.G _{mat}	Shear modulus (Pa)	SMT.m	WingboxSizing.m	Auto-computed
<i>L_{min,AI}</i>	loc.AI.L _{min} = 2 mm	AI minimum skin gauge (m)	SMT.m	WingboxSizing.m	Auto-computed
<i>tw_{min,AI}</i>	loc.AI.L _{web} _{min} = 3	AI minimum web gauge (m)	SMT.m	WingboxSizing.m	Auto-computed
<i>Ac_{min,AI}</i>	loc.AI.A _{cap} _{min} = 1e-4	AI minimum spar cap area (m ²)	SMT.m	WingboxSizing.m	Auto-computed
<i>G_{CF}</i>	loc.CF.E = 59.8 GPa	CFRP Young's modulus (E8 rule) (Pa)	SMT.m	WingboxSizing.m	Auto-computed
<i>G_{CF}</i>	loc.CF.G = 18.8 GPa	CFRP shear modulus (Pa)	SMT.m	WingboxSizing.m	Auto-computed
<i>nu_{CF}</i>	loc.CF.nu = 0.33	CFRP Poisson's ratio	SMT.m	WingboxSizing.m	Auto-computed
<i>rho_{CF}</i>	loc.CF.rho = 1550	CFRP density (kg/m ³)	SMT.m	WingboxSizing.m	Auto-computed
<i>sig_{CF}</i>	loc.CF.sig _{all} = 220 MPa	CFRP allowable (post-BVID) (Pa)	SMT.m	WingboxSizing.m	Auto-computed
<i>tau_{CF}</i>	loc.CF.tau _{all} = 110 MPa	CFRP allowable shear (Pa)	SMT.m	WingboxSizing.m	Auto-computed
<i>b_{rib}</i>	loc.rib.spacing = 0.60	Rib pitch / skin panel width (m)	SMT.m	WingboxSizing.m	Auto-computed
<i>k</i>	loc.k.buckle = 4.0	Simply-supported plate buckling coeff.	SMT.m	WingboxSizing.m	Auto-computed

